

Life in grassland

Despite centuries of human disturbance, grassland supports some of the largest concentrations of animal life on earth.

Survival in grassland habitats is far from easy, however: aside from the lack of shelter and plant diversity, there are hazards such as drought and fire to contend with. Added to this is the ever-present risk of attack by some of the world's fastest and most powerful predators.

Herding

Life in open grassland is often dangerous because there are few places to hide. To increase their chances of survival, many large plant eaters live in herds. This makes it more difficult for predators to attack, because while most members of the herd are eating, some are always on the lookout for danger.

Today's largest herds are found on Africa's plains. Here, migrating wildebeest can form herds over a quarter of a million strong, and 25 miles (40 km) long, although even these herds are small compared with some that existed in the past. During the 19th century, springbok herds in southern Africa sometimes contained more than 10 million animals. In North America, bison herds probably reached similar sizes before hunting brought the species to the edge of extinction.

Life in herds does have its problems, one of which is the risk that an animal might wander off and become lost. Most herding species have scent glands on their hooves so that if an animal becomes isolated it can follow the scent tracks to rejoin the herd. Another problem is giving birth. To prevent their young from being trampled or attacked, many grazers give birth in cover, and rejoin the herd a few weeks later. Some, however, are born in the open and have to be able to keep up with the herd when they are just a few hours old.



Movement

In grassland and savanna, there is a premium on speed. It is no accident that the world's fastest land animals, such as the cheetah and the pronghorn, are found in this habitat. Natural selection favours predators that are fast enough to catch food, and prey animals that are fast enough to escape.

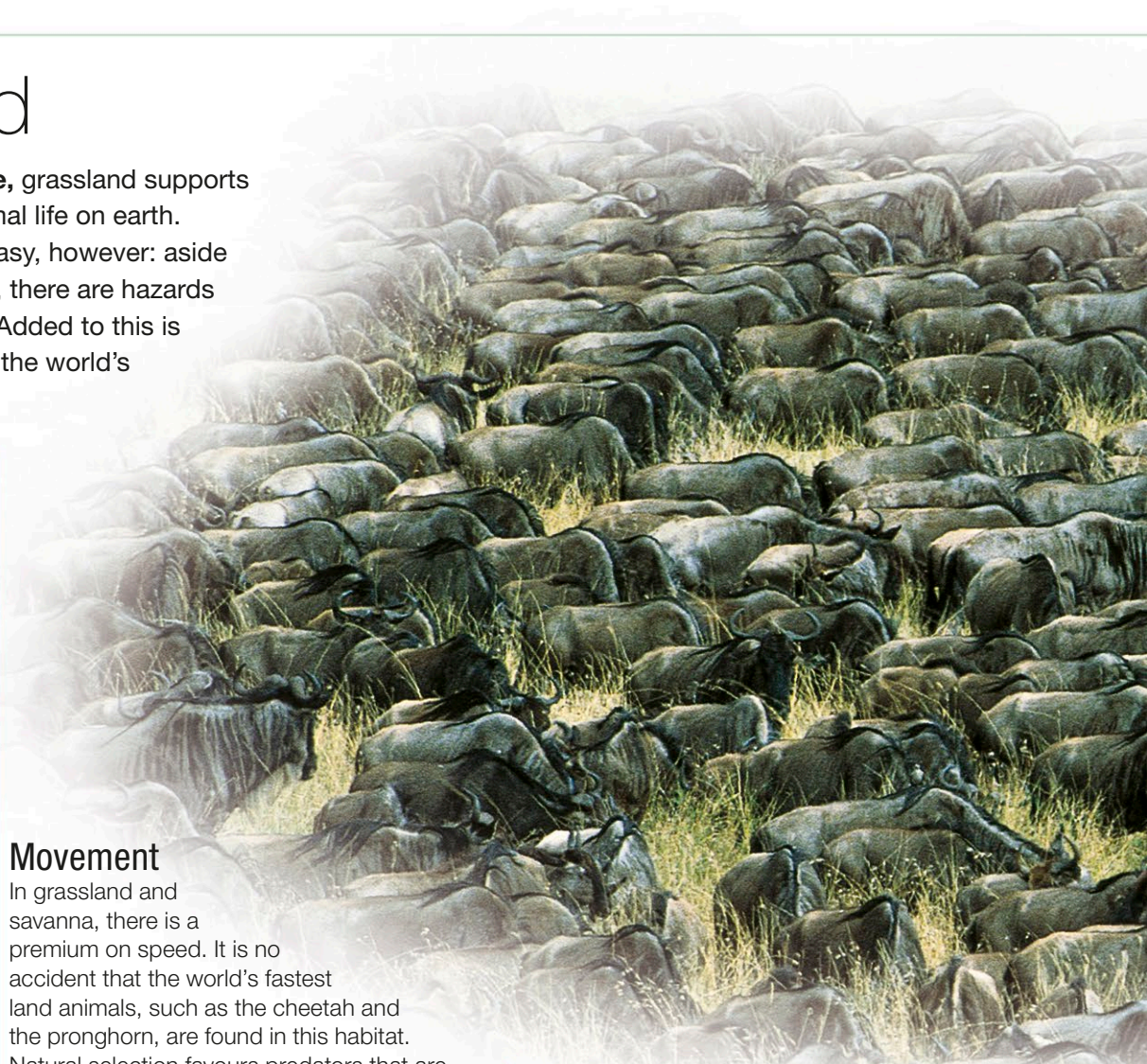
Most of the fast runners are mammals; however, grassland also has nature's fastest-running birds, including ostriches, rheas, and emus—giant species that have lost the ability to fly. These birds can reach speeds of up to 44 mph (70 kph). More importantly, they are able to maintain such speeds for up to 30 minutes—long enough to outrun most of their enemies unless a predator launches an attack from a very close range.

Despite the many fast runners, grassland life often appears tranquil. This is because running is extremely energy-intensive, and animals run only when they absolutely have to. Prey animals have invisible "security thresholds" that vary according to the threat they face. For example, gazelles often let lions approach to within about 650 ft (200 m) because they are instinctively aware that lions that are visible at this distance are unlikely to be stalking prey.

A solitary cheetah, on the other hand, will send a gazelle herd sprinting, even if it is seen to be 4 times farther away.

FLIGHTLESS BIRDS

Standing guard over its chicks, a greater rhea watches for danger. Like other flightless birds that live in grassland, it uses its height and large eyes to spot potential predators at a distance so that it has time to run for safety.



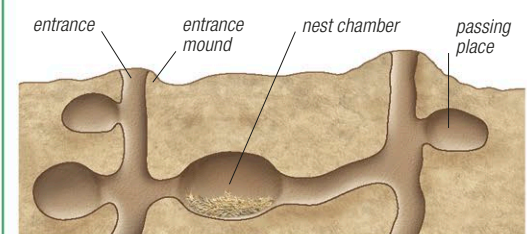
Living underground

Some grassland animals find safety not by running away but by retreating into burrows below ground. There, they can stay out of reach of most predators and find some protection from the worst of the elements. Subterranean animals include a wide variety of species, ranging from mammals to insects. Some animals, particularly snakes, do not excavate their own burrows; instead, they adopt existing ones. The largest burrows, made by African aardvarks, are big enough to accommodate a person, and are a serious hazard to vehicles; the most extensive are made by prairie dogs and other rodents. Before farming became widespread in North



MAKING A HOME BELOW GROUND

A black-tailed prairie dog collects grass to line a nesting chamber. The colony, or town, consists of tunnels up to 16 ft (5 m) deep.



PRAIRIE-DOG BURROWS